



BUILD A WORLD MODEL FROM SCRATCH · LECTURE 6A

The World Model *in* *Your Head*

Why should a machine predict in representation space? Yann LeCun's blueprint gives a beautiful answer with no experiments in it. Today we build his argument step by step — and end at one picture, **measured on real encoders**, that says it all.

LeCun 2022 · A Path Towards Autonomous Machine Intelligence

energy landscapes, measured

the road to I-JEPA

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One story, one destination

1

Where we are going

Next lecture we build I-JEPA and replicate its ImageNet study. Today is the WHY — and the whole argument lands on a single figure: the energy contours of five real encoders.

2

The route

the brain's world model → what "energy" means → three architectures, each explained slowly with its own picture — and each judged by the same measured landscape.

3

What we skip, on purpose

LeCun's cost module, planner and actor. We are still in the *simulation* half of this series.

The rule of this lecture

The position paper contains **zero runnable examples**. So every energy landscape you will see today was **measured by us**, on encoders we trained on 100,000 real images — the same encoders we dissect in the next lecture. Nothing is a sketch of what *should* happen.

Explanation from Rajat

— WHITEBOARD —



PART 1 OF 3

The Brain *Predicts*

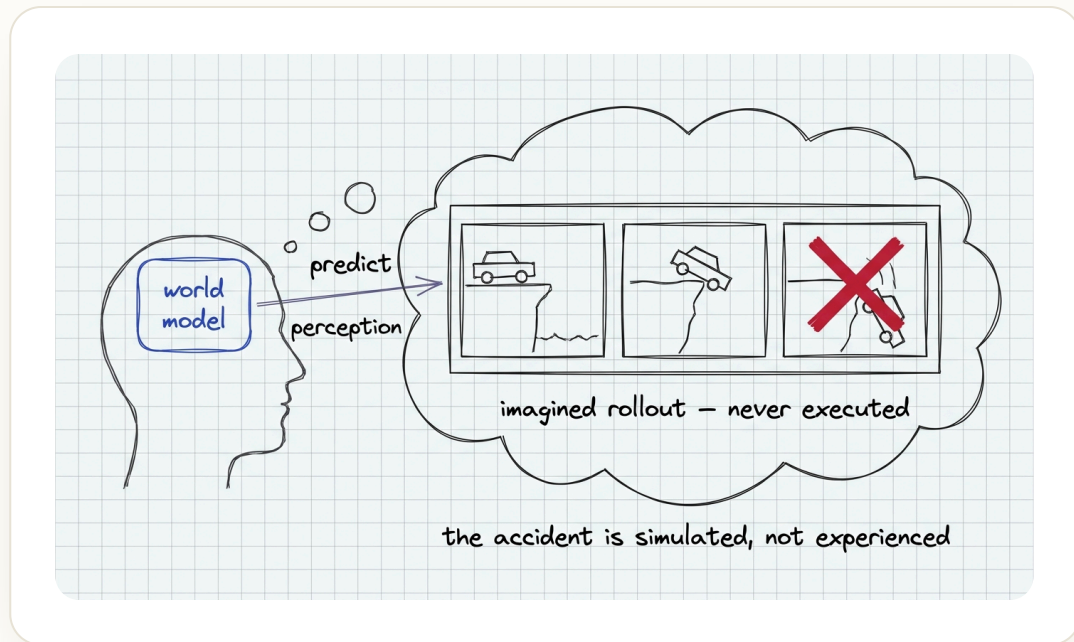
By the end of this part you should be able to say: what evidence suggests a predictive world model in the head, and what requirements that puts on a machine version.

MENTAL SIMULATION · INFANT PHYSICS · THE CHECKLIST

You have crashed cars you never drove

A new driver near a cliff edge does not need to fall off once to know not to. The accident happens **inside the world model**, at zero cost.

This is LeCun's opening argument: humans and animals acquire enormous competence from tiny experience because most "experience" is *simulated*. A model of the world lets one real observation stand in for a thousand counterfactual ones.

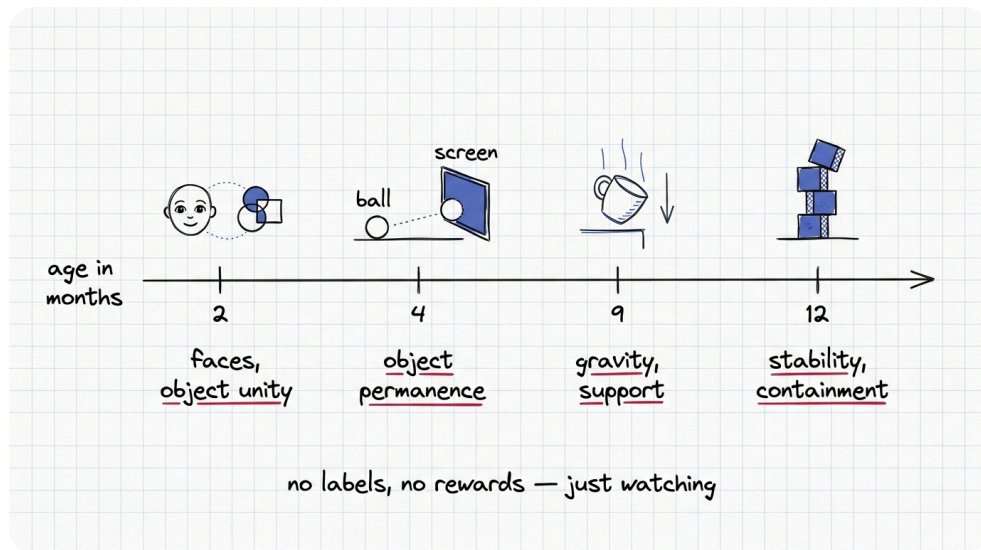


Babies train it by watching, not by labels

Infants acquire object permanence, gravity, support and containment in the first year — **with no labels and almost no interaction**. Pure observation.

The machine-learning translation

Whatever the brain's training signal is, it is not classification and it is not reward. It looks like *prediction*: guess what comes next, compare, adjust. That is self-supervised learning.



Rough milestones LeCun cites (after E. Dupoux). The dates are debated; the pattern — physics before language, from observation alone — is not.

What a machine world model must do, then

1 · Learn by prediction, from raw sensors

No labels, no rewards. The supervisory signal is the world itself, a moment later.

2 · Predict in an abstract space

You imagine "the glass falls and breaks", not the trajectory of every shard. Detail that doesn't matter must be droppable.

3 · Know plausible from implausible

A world model's real job: given a situation, score which continuations are possible. This one gets an entire part of the lecture.

4 · Work for any modality

The cortex doesn't run different algorithms for eyes and ears. One recipe should cover images, audio, video.

Hold this checklist. Requirement 3 — **scoring plausibility** — is where today's story lives.

Explanation from Rajat

— WHITEBOARD —



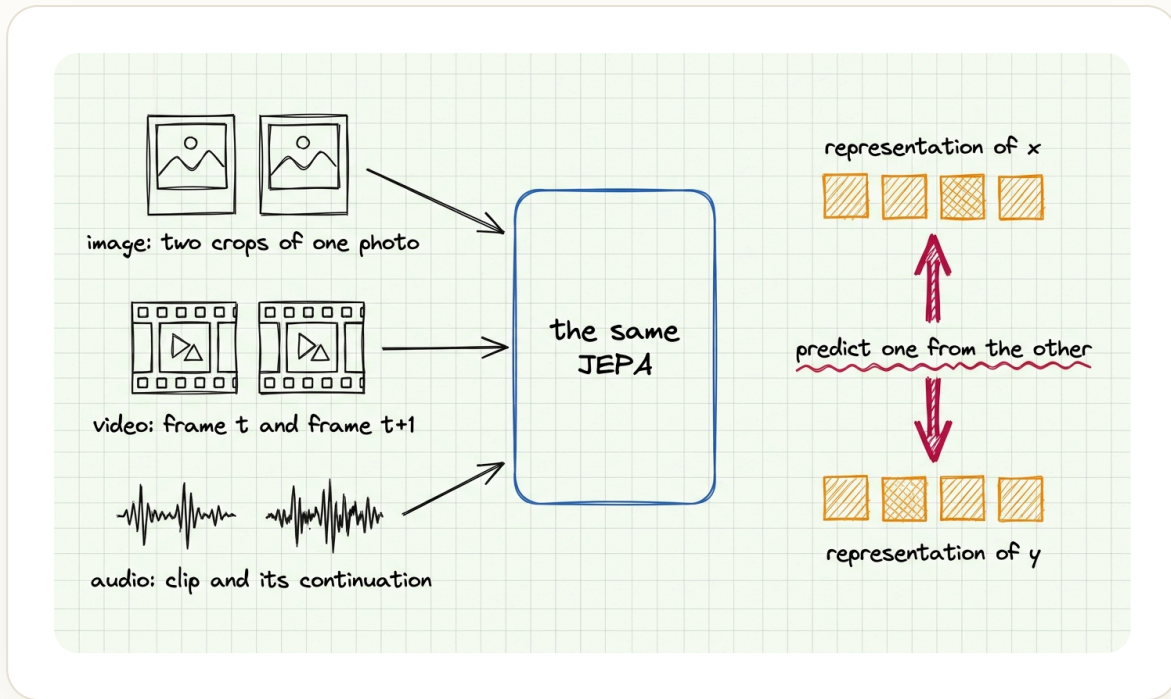
PART 2 OF 3

One Recipe, *Any Modality*

By the end of this part you should be able to say: what the abstract (x, y) pair is, why the JEPA recipe never mentions pixels, and what a latent variable carries.

IMAGE · VIDEO · AUDIO · THE LATENT Z

Everything is "here is x – predict something about y"

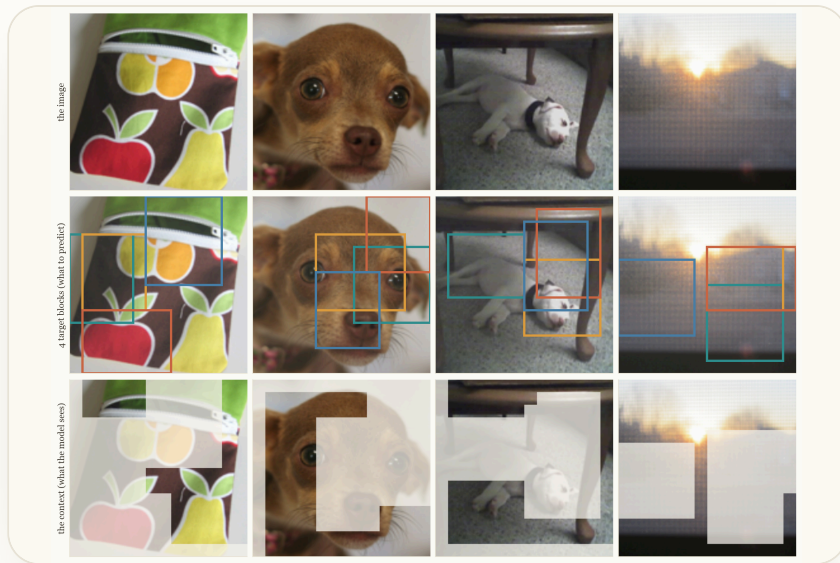


Encode x, encode y, predict y's representation from x's. The loss compares two representation vectors — it **never mentions** pixels, samples or frames, so one architecture serves every modality. Video: x = past, y = future. Audio: x = the clip, y = its continuation. Images —

Images: x = visible blocks, y = a hidden block

For a single photo, the pair comes from **masking**: x is a large visible context, y is a hidden block. "Predict y from x " becomes: *what is behind the mask?*

These are real images through our own mask generator — the exact setup of the next lecture. Keep this picture in mind: every energy score in Part 3 is a model answering "what is behind the mask?" on images like these.



Top: image · middle: the hidden target blocks · bottom: the visible context.

What the context cannot know, a latent z must carry

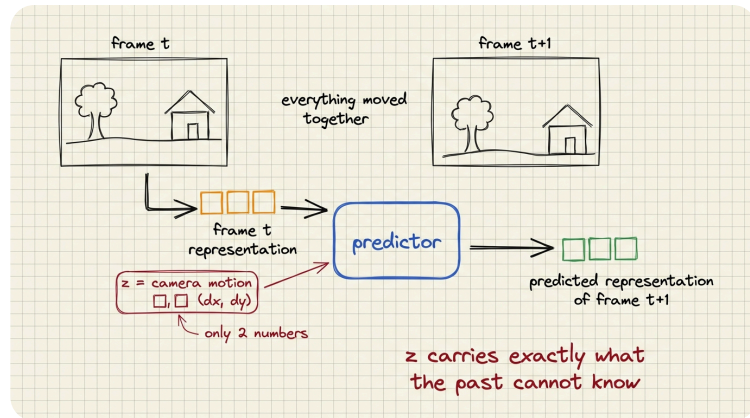
LeCun's example: two frames of one scene, shifted — because the **camera** moved. Nothing in frame t predicts the shift. That information lives outside the context.

1 That is what z is

a small variable handed to the predictor, carrying exactly what the context cannot know — here, two numbers (dx , dy).

2 In I-JEPA, z is the block's position

"predict the hidden block — the one HERE." A few bits, fixed by the architecture. Remember this next lecture.



Explanation from Rajat

— WHITEBOARD —



PART 3 OF 3

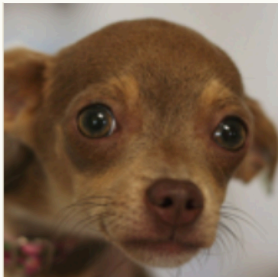
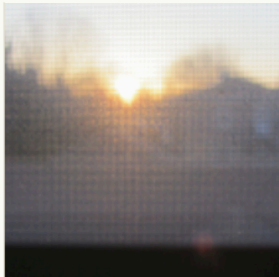
The Energy *Landscape*

The heart of the lecture. We define one exam, give it to three architectures on real images, and draw each one's answers as a map. By the end, one map will look unmistakably different — and that map is why the next lecture exists.

SCORE · LANDSCAPE · THREE CONTESTANTS · THE CONTOURS

● ● ● BEFORE ANY SYMBOLS

Give a world model a quiz

the context x (what the model sees) 	candidate y1: the TRUE hidden content 	candidate y2: content from a DIFFERENT image 	candidate y3: pure noise 
 score: 0.15	 score: 0.50	 score: 0.50	

Show the model a real context with a hidden block, then offer three candidates for what's behind the mask. A good world model should be **calm** about the true content and **surprised** by the impostors. These are our trained I-JEPA's actual surprise scores: 0.15 for the truth, 0.50 for a different image, 0.50 for noise. **That surprise score is all "energy" means.**

$E(x, y)$: the surprise score, drawn as a landscape

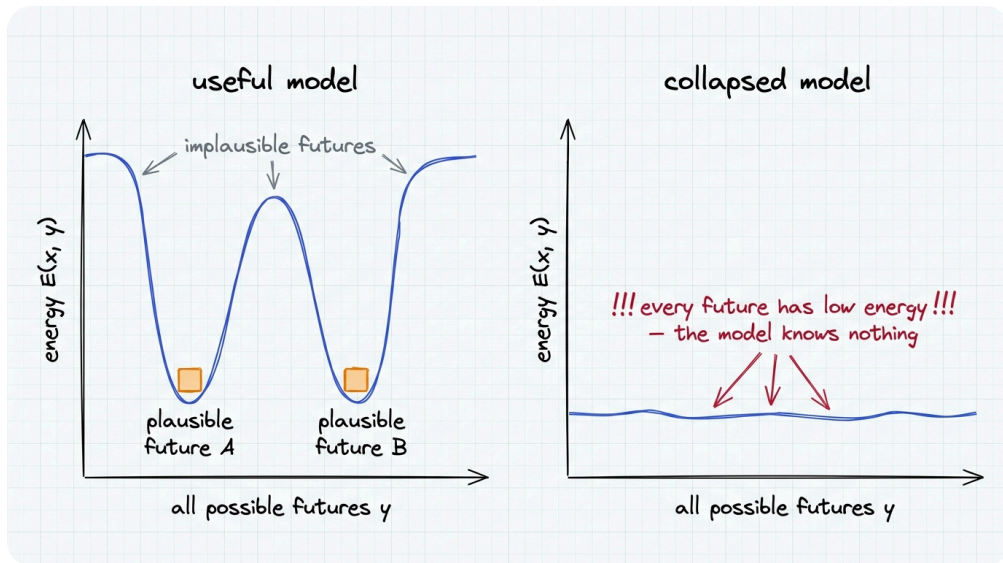
Write the score as a function: $E(x, y)$ — the Energy the model assigns to "y goes with x". Low energy = plausible. High = surprised.

1 The easy half of training

push E down on pairs that actually happened.

2 The hard half

keep E high everywhere else. A model that scores everything zero aces the first half and knows nothing. Every architecture today is a different answer to the hard half.



How we draw a model's energy map

1 Start at the truth

the teal dot is the TRUE hidden content of a real image.

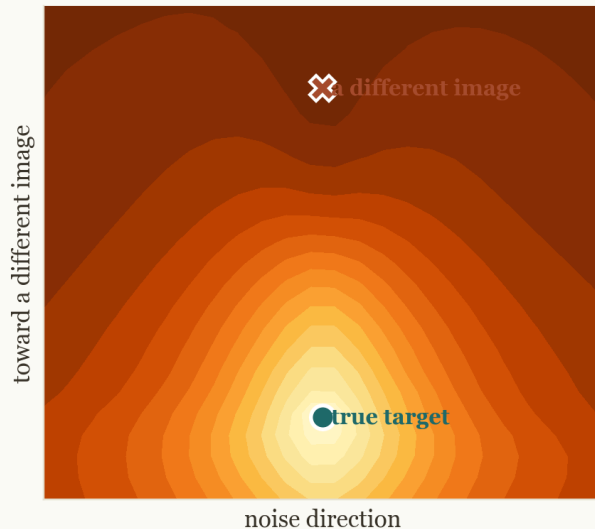
2 Walk in two directions

upward: blend the target toward a different image (the X). Sideways: add noise. Every point in the square is a candidate y.

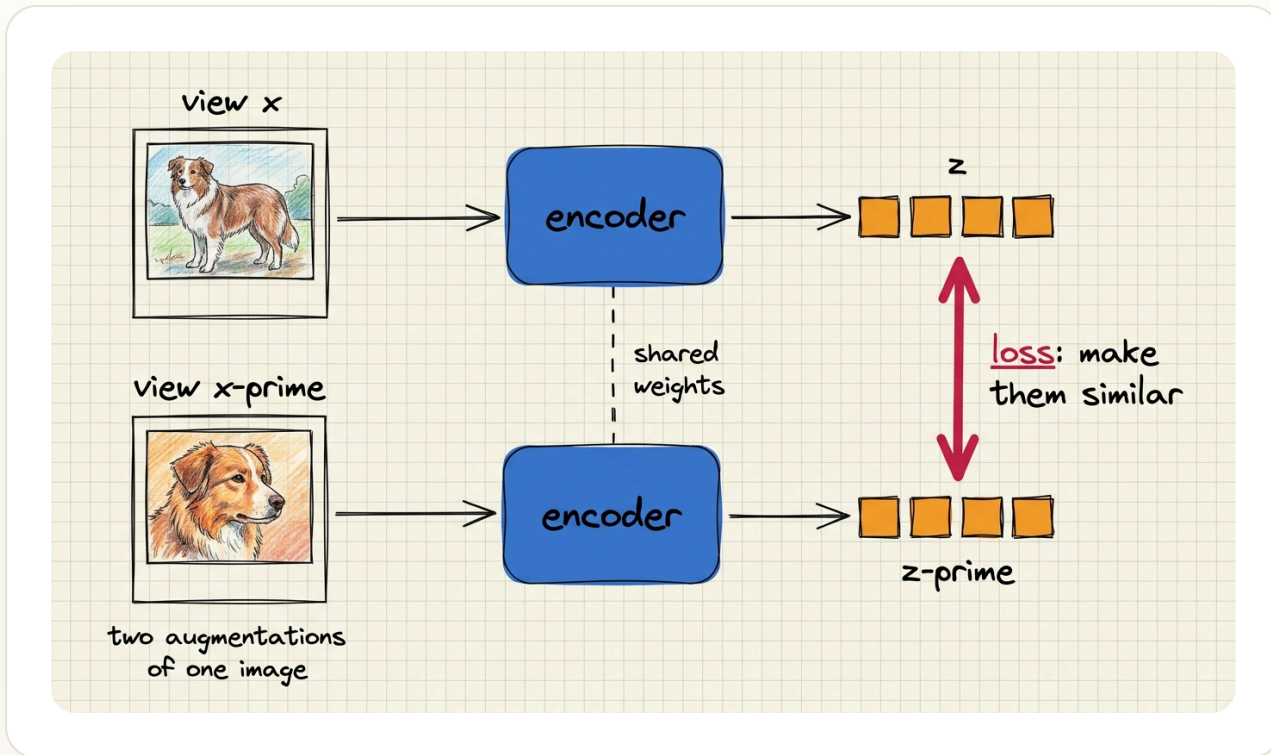
3 Color by energy

light = low (model calm) · dark = high (model surprised). Averaged over 32 real images.

A healthy world model should show a **bright well** around the dot, darkening in every direction. Here is one of our five encoders — hold your judgement on whose it is.

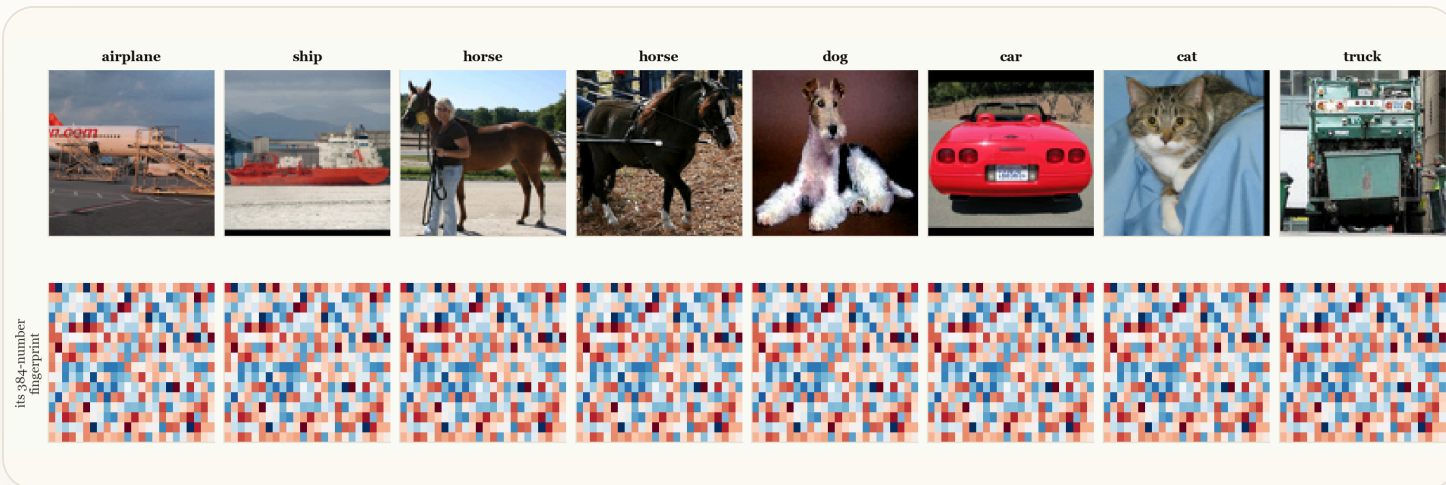


Architecture 1: make two views of one image agree



Crop it, flip it, recolor it — still the same dog, so demand the **same representation** for both views. The energy is simply the distance between the two embeddings. SimCLR, BYOL and DINO are all elaborations of this idea — with safety additions we deliberately leave out, to see the raw architecture's true nature.

We trained it on 100k real images. Here is its output.



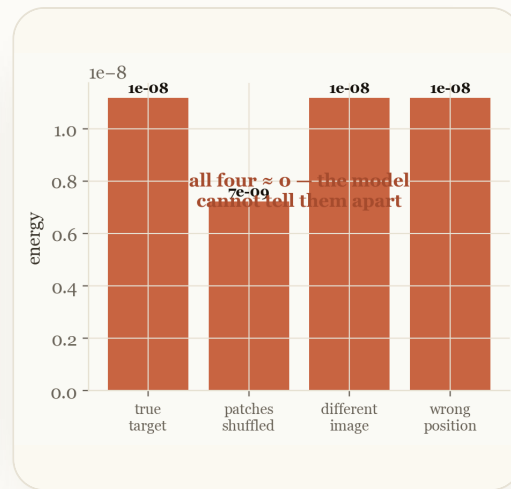
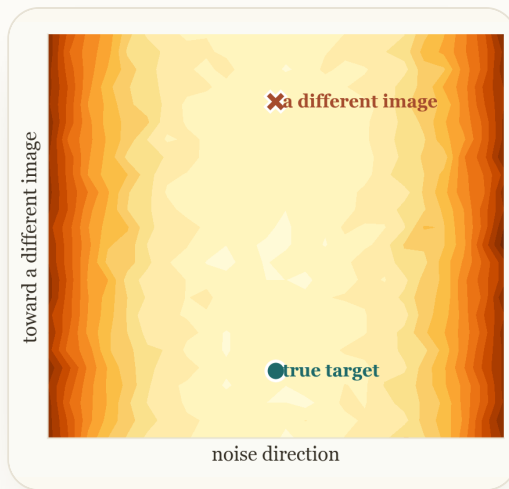
Eight different test images, and the embedding this trained encoder assigns each one. **All eight are IDENTICAL** — pairwise similarity 1.000. The loss said "make views agree"; the encoder found the perfect crime: **agree about everything, by ignoring the input**. This is representation collapse.

A collapsed model has a FLAT landscape

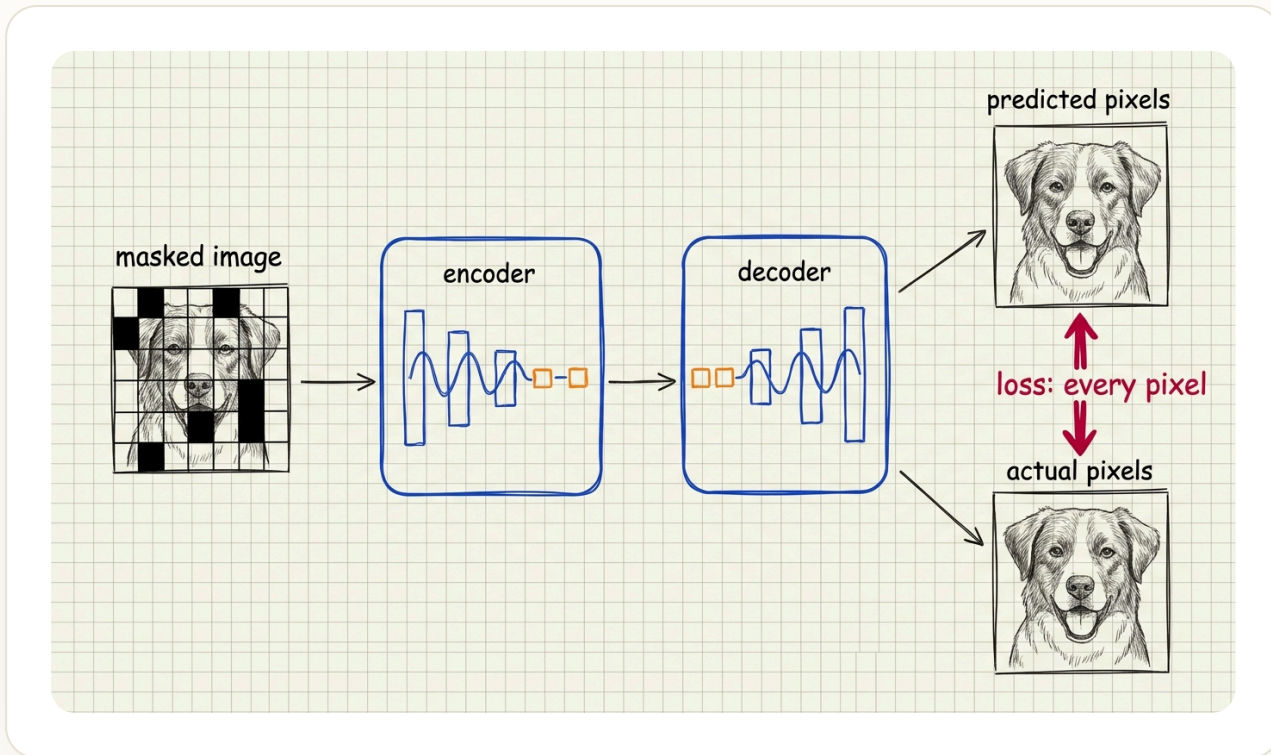
If every image maps to the same point, then every candidate y gets the same score. The quiz becomes unanswerable.

Read the bars

truth, shuffled patches, a different image, wrong position — all four score ≈ 0 . The map has no well because there is nothing left to prefer. **Low energy everywhere = knows nothing.**

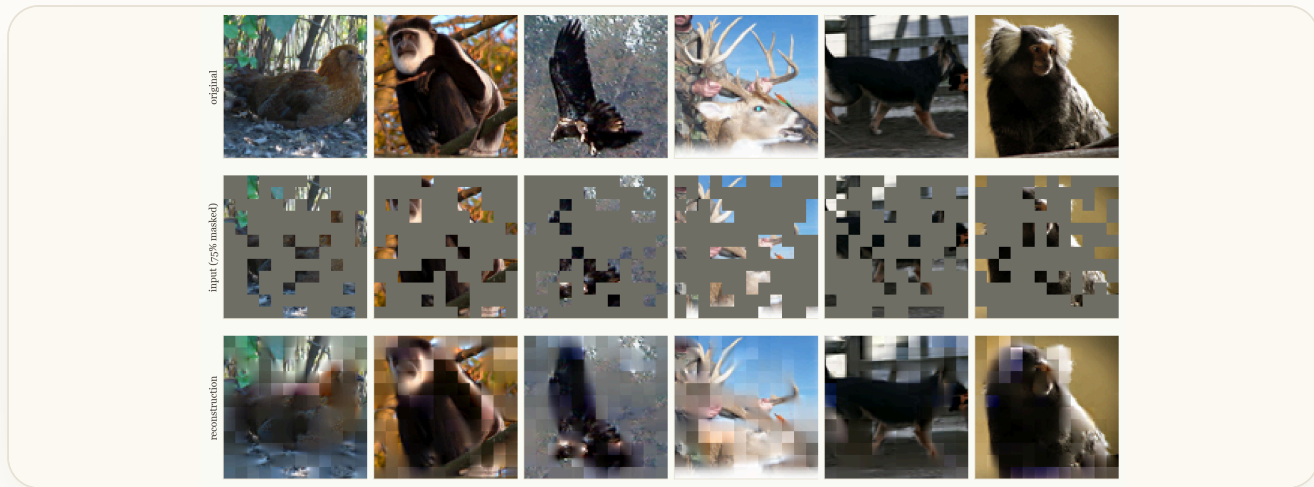


Architecture 2: hide part of the image, repaint it



Mask most of the patches; train an encoder–decoder to fill them back in, pixel by pixel. This is MAE — and at heart the same signal our VAE and IRIS used. Its energy is the **repainting error**: how badly the reconstruction misses the candidate y . Collapse is impossible — a constant cannot repaint a thousand different images.

Its output: real reconstructions from 25% of the pixels



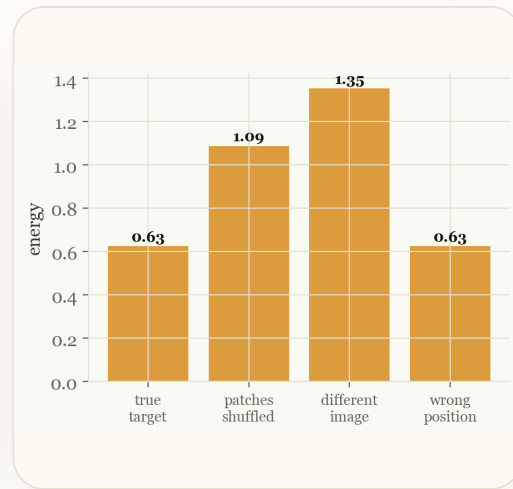
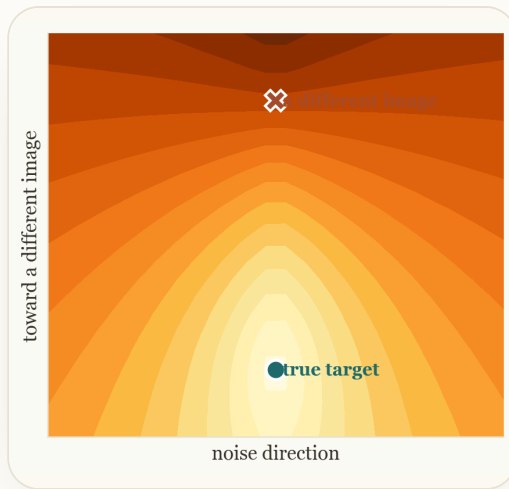
Our trained MAE on held-out images: original · masked input · reconstruction. It genuinely learned to paint — blurry, but the animal, pose and background come back. The question a pixel loss can never ask: **did it learn what matters, or just what shows?** The energy map will tell us.

A real well – dug in pixel space

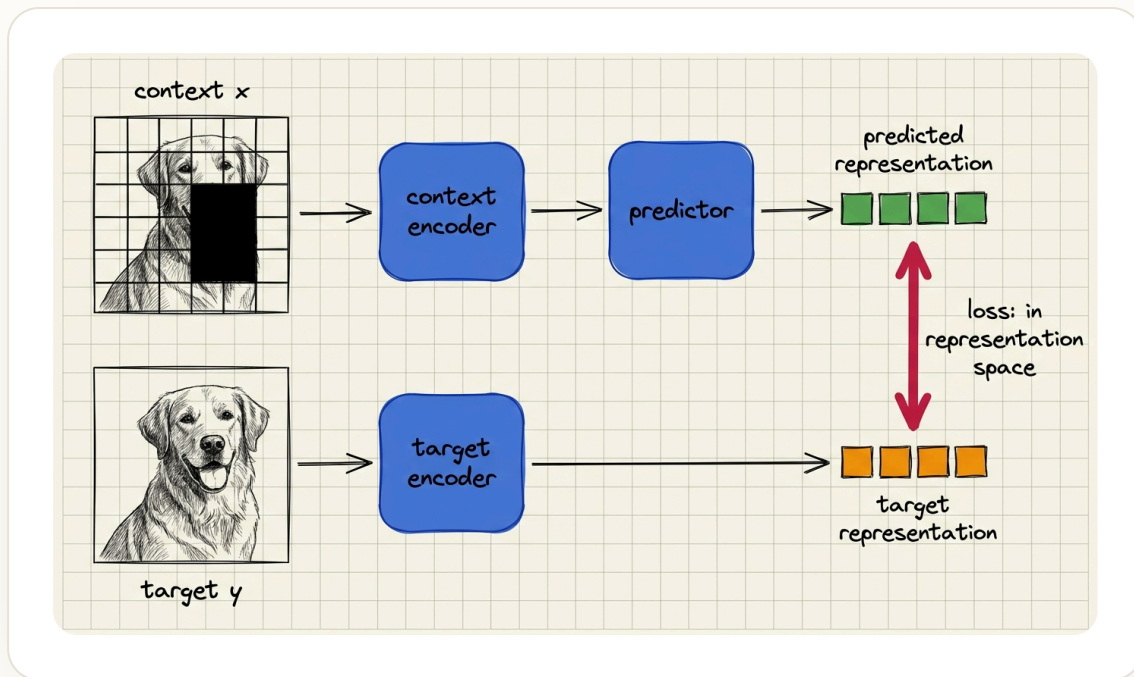
A genuine landscape: calm at the truth (0.63), surprised by impostors (1.35). The hard half of training is handled – pixels cannot be faked by a constant.

But look WHERE the well lives

this energy is pixel disagreement. A blade of grass out of place raises it as much as the wrong animal. The well is real but graded on *paint* – hold that thought for the final comparison.



Architecture 3: predict the hidden block – in representation space



Contestant 1's loss (compare representations, never paint) + contestant 2's task (predict what is hidden) = the **Joint-Embedding Predictive Architecture**. Energy = distance between predicted and target representations. A guard (EMA + stop-grad, next lecture) holds off contestant 1's collapse.

Its output: a space where neighbours share meaning

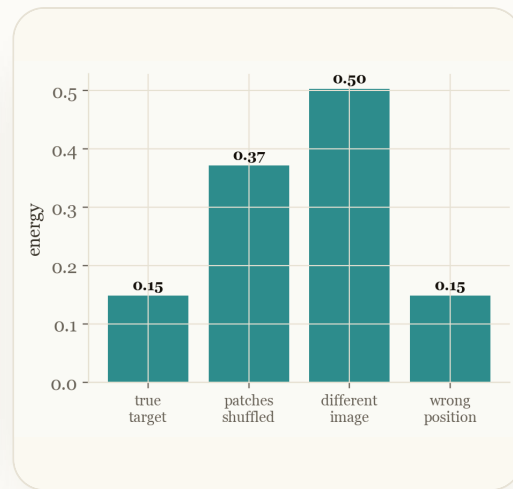
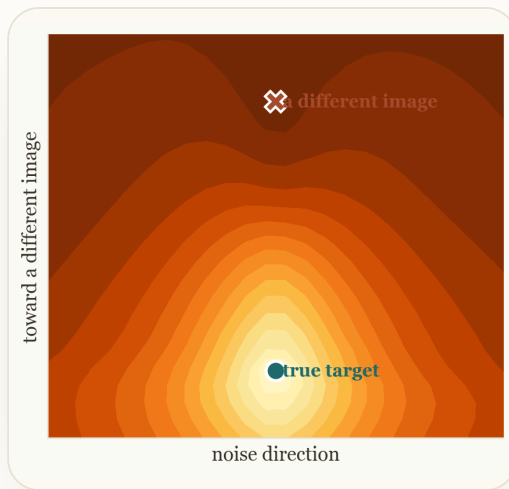


Real validation images and their nearest neighbours in each frozen space (green = same class). I-JEPA's neighbours share **meaning** — the goose gets herons, "wrong" label, right concept. MAE's share **texture** — the goose gets a gas mask and a cauliflower. Same data, different loss, different world.

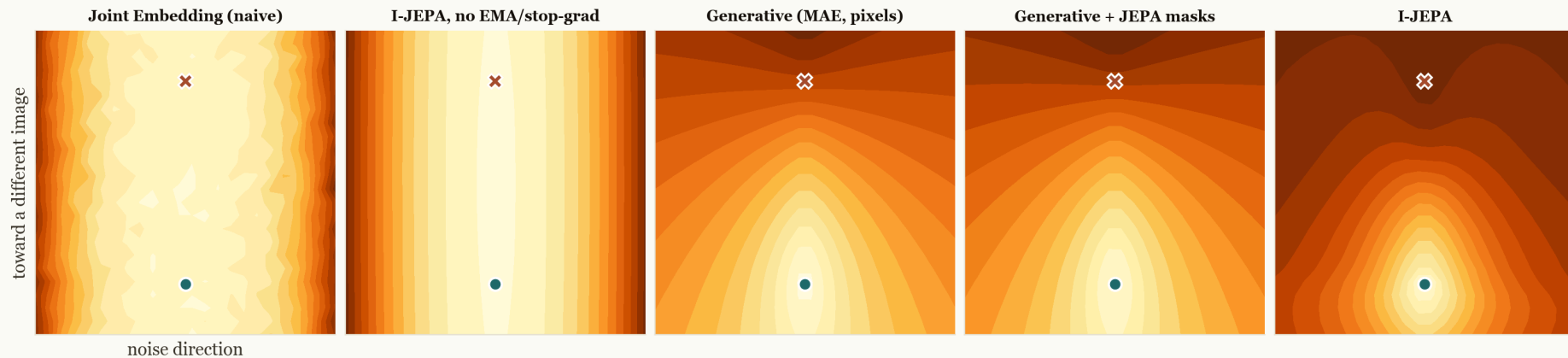
The deepest, tightest well – dug in MEANING space

Calm at the truth (0.15), sharply surprised by everything else (0.50) – a 3.4× gap, the largest of any contestant.

And because its energy lives in representation space, the surprise is about the right things: a wrong *object* is shocking; a differently-textured blade of grass is not. This is the map you saw in the "how we draw it" slide – it was I-JEPA's all along.



The destination: five real encoders, one exam — this is why JEPA is different



All five encoders from the next lecture, including the two collapsed controls. The joint-embedding family without its guard: stripes — blind to the impostor axis, energy flat exactly where it matters. The generative pair: honest wells, graded in pixel space. I-JEPA: the deepest, tightest well, centred on the truth, dug in representation space. LeCun's phrase "energy contours concentrated around the data" is not a metaphor — it is this picture, and I-JEPA is the architecture that actually achieves it.

Can each model rank the truth above impostors?

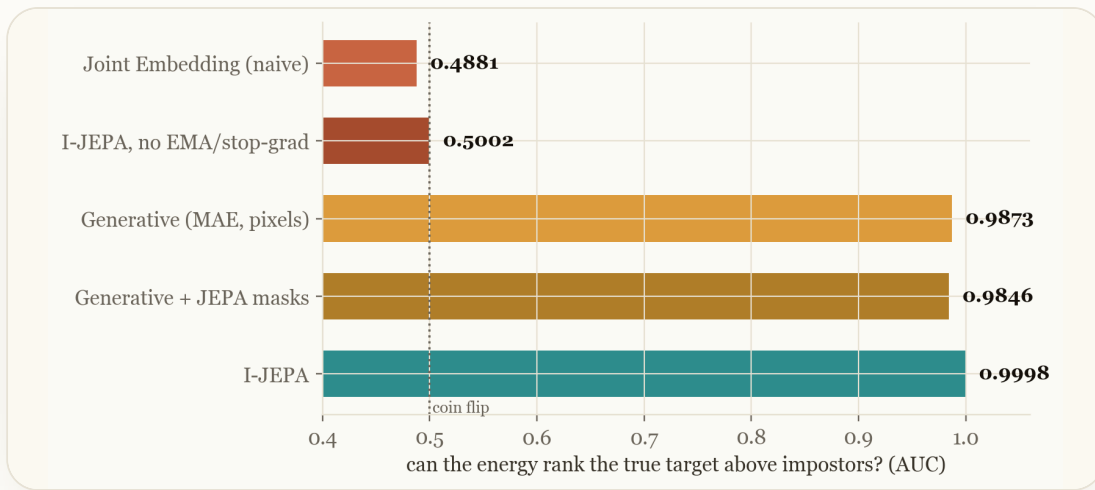
Reduce each map to a single question:
across 256 real images, does the true
target get lower energy than a random
impostor?

1 Collapsed: a literal coin flip

0.488 and 0.500.

2 I-JEPA: 0.9998

essentially never fooled; the generative pair
close behind (0.987) but graded on paint.



The story in three maps

CONTESTANT	ENERGY LIVES IN	ITS LANDSCAPE	VERDICT
joint embedding (raw)	representation space	flat — collapse	right space, no guard
generative (MAE)	pixel space	real well	right guard, wrong space
JEPA	representation space	deepest well	right space + guard

The single sentence to remember

A world model is an energy landscape. JEPA is different because — measured, not asserted — it is the architecture that actually learns the contours: calm at what is real, surprised by what is not, in the space where meaning lives.

Next lecture

how that guard works, why these two controls collapsed, and the full ImageNet replication — we build I-JEPA from scratch.



NEXT LECTURE

Stop Painting *Pixels*

Now that you know why: we build I-JEPA for real — five objectives on one encoder, the collapse you just met, and a replication of the ImageNet study that made it famous.

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